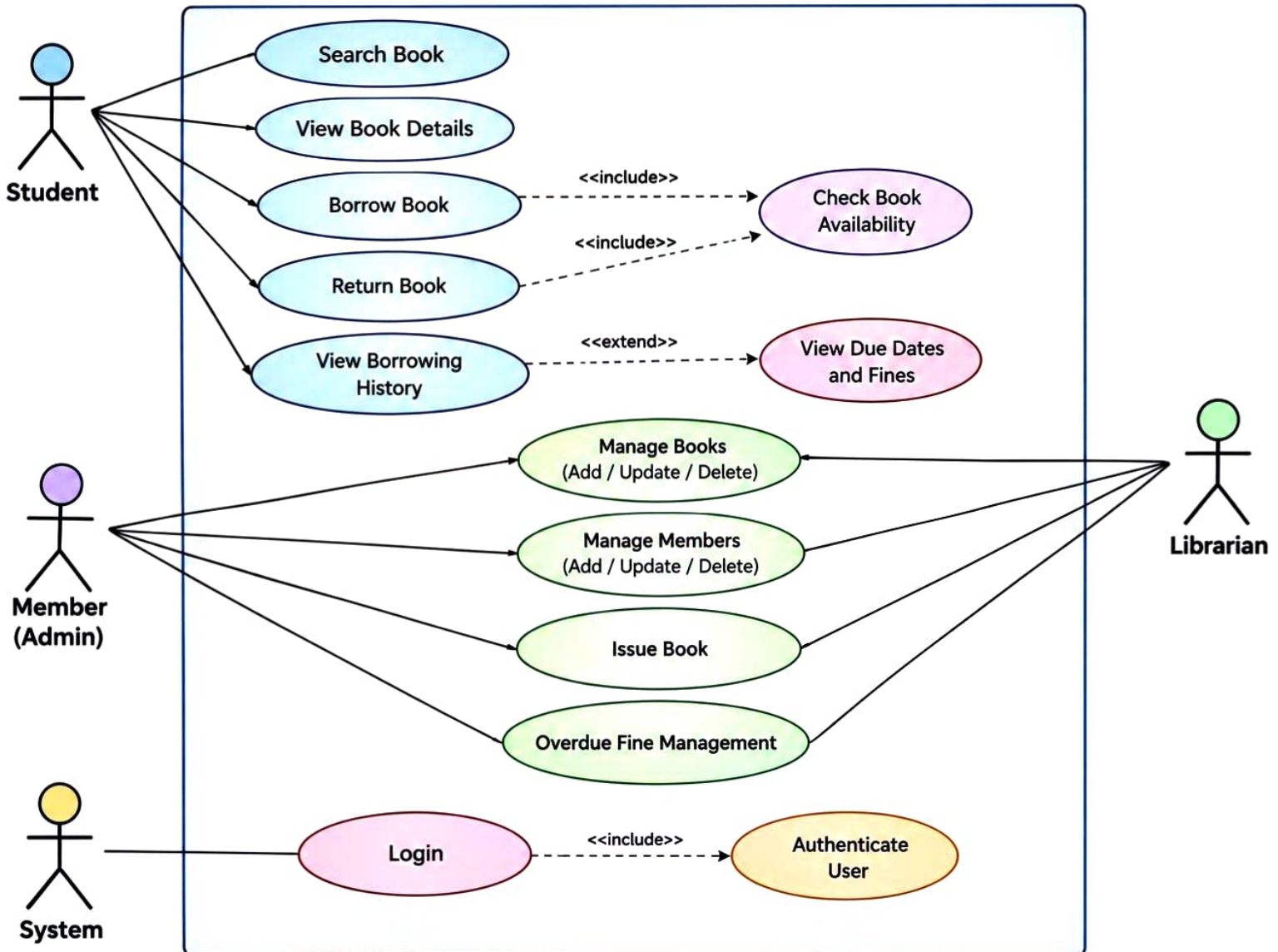
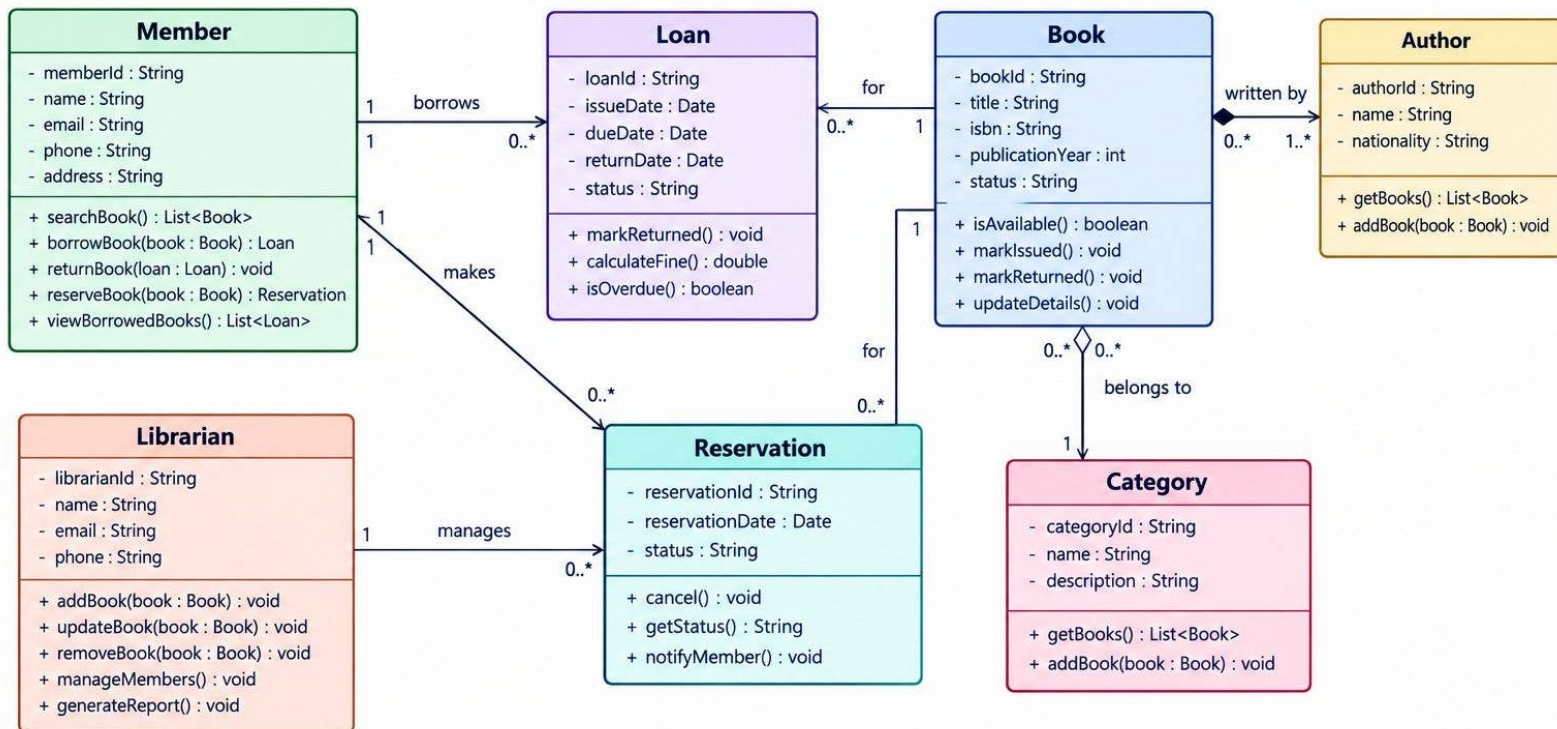


Library Management System

(Use Case Diagram)



Class Diagram – Library Management System



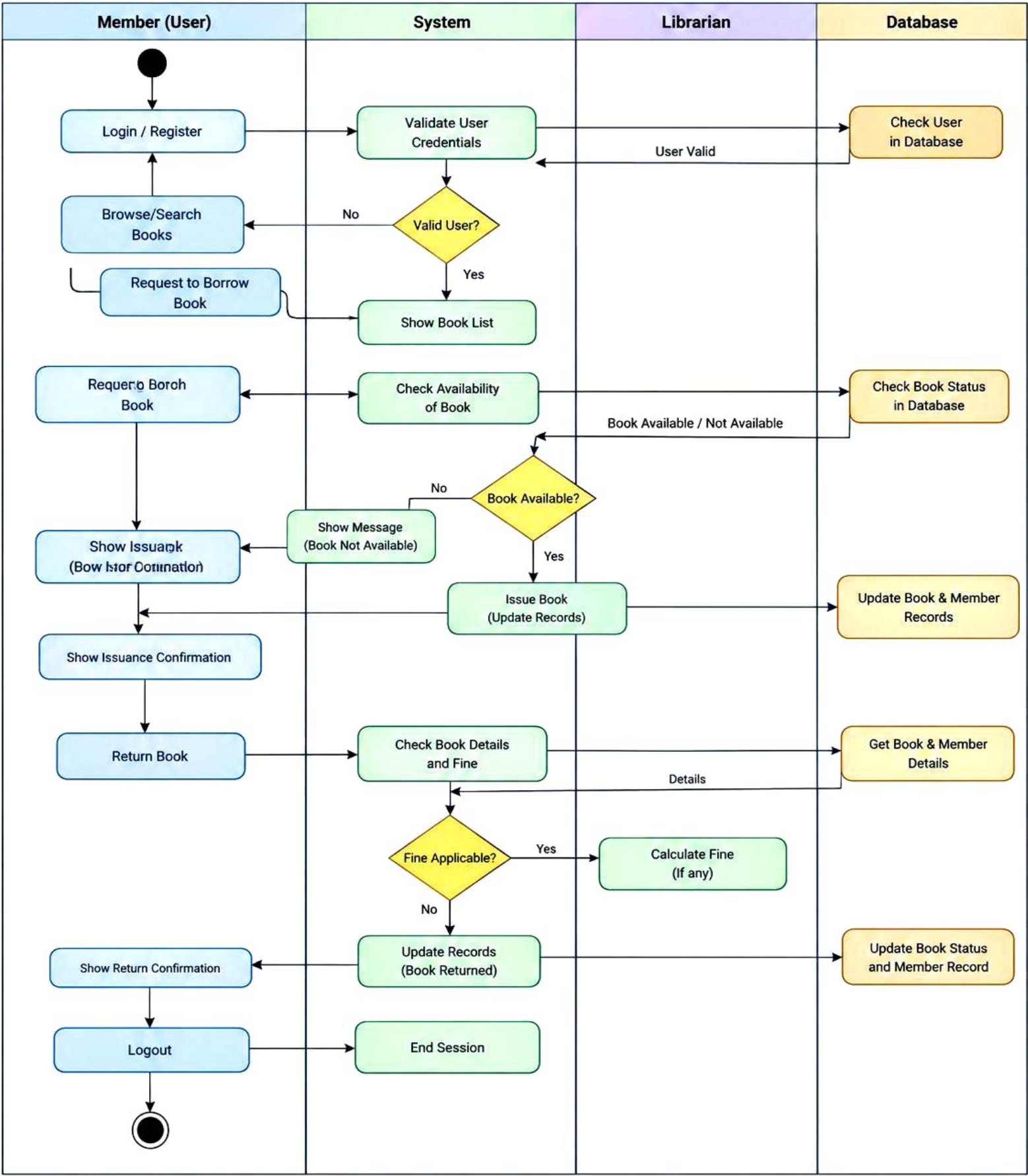
Legend:

— Association ◇ Aggregation ◆ Composition ▷ Generalization (Inheritance)

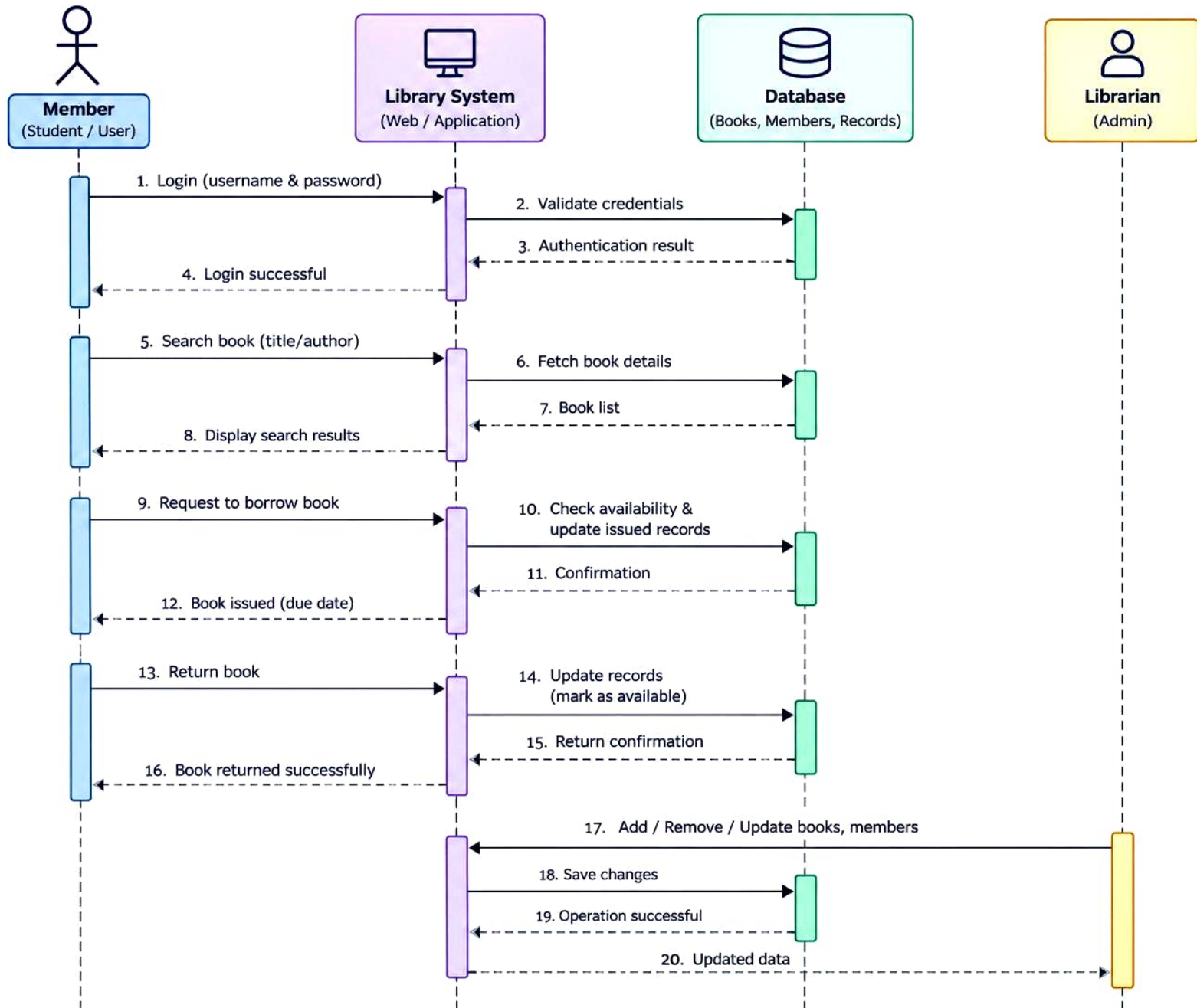
Note:

1. A Member can borrow multiple books (0..*), but each loan is for one book (1).
2. A Book can have multiple reservations and loans.
3. A Book belongs to one category, and a category can have many books.
4. A Book can be written by one or more authors, and an author can write many books.
5. Librarian manages books, members and reservations.

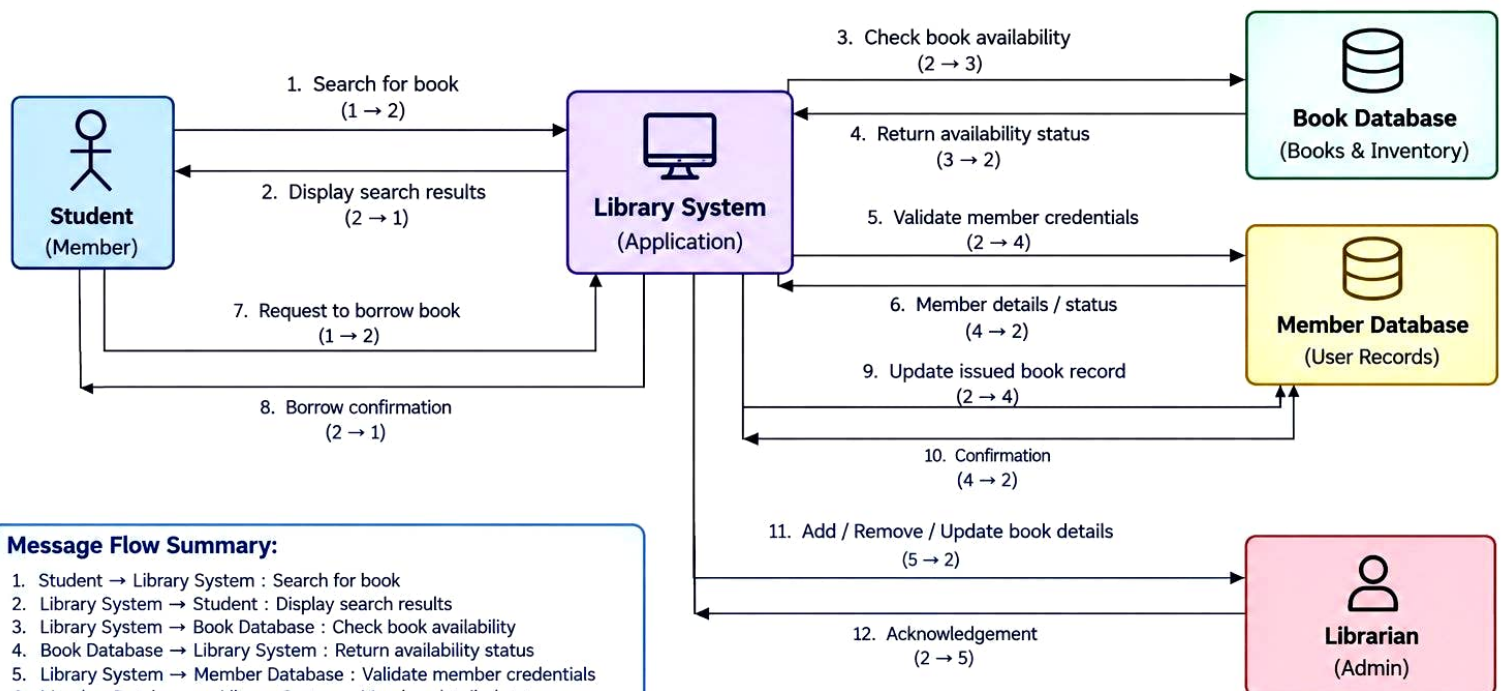
Activity Diagram – Library Management System



Sequence Diagram – Library Management System



Communication Diagram – Library Management System



Message Flow Summary:

1. Student → Library System : Search for book
2. Library System → Student : Display search results
3. Library System → Book Database : Check book availability
4. Book Database → Library System : Return availability status
5. Library System → Member Database : Validate member credentials
6. Member Database → Library System : Member details / status
7. Student → Library System : Request to borrow book
8. Library System → Student : Borrow confirmation
9. Library System → Member Database : Update issued book record
10. Member Database → Library System : Confirmation
11. Librarian → Library System : Add / Remove / Update book details
12. Library System → Librarian : Acknowledgement

Statechart Diagram – Library Management System

